Highlights

A Multi-Layer Framework for Filtering, Merit Classification, Sentence Segmentation, and Named-Entity Extraction in Brazilian Criminal Labor-Law Decisions

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- Multi-layer filtering for noisy, heterogeneous legal corpora.
- Task-specific normalization improves classification and NER compatibility.
- Conviction-focused segmentation and legal-domain NER for structured extraction.
- Reproducible benchmark design for Portuguese criminal labor-law decisions.

A Multi-Layer Framework for Filtering, Merit Classification, Sentence Segmentation, and Named-Entity Extraction in Brazilian Criminal Labor-Law Decisions

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ABSTRACT

This paper presents a multi-layer legal NLP framework for extracting structured information from Brazilian criminal labor-law judicial decisions related to modern slavery. Starting from a heterogeneous corpus (2,581 documents), the framework applies: (i) document-type filtering to retain only final decisions, (ii) legal-text normalization with task-specific outputs for classification and NER, (iii) penal-merit classification using Legal-BERT embeddings with classical classifiers, (iv) LLM-assisted sentence segmentation for conviction-focused documents, and (v) legal-domain NER for entity extraction. We describe the full pipeline, evaluation protocol, and error analysis, and discuss practical deployment constraints, reproducibility, and ethical/legal considerations. Results indicate that layered filtering and task-oriented preprocessing substantially improve downstream extraction quality in low-resource legal settings.

1. Introduction

1.1. Problem context and motivation

- Explain the supply-chain transparency challenge in modern slavery investigations.
- Justify why judicial decisions are high-value evidence sources.

1.2. Research gap

- Existing legal NLP pipelines are often single-stage and not robust to document heterogeneity.
- Limited resources for Portuguese criminal labor-law datasets and benchmarks.

1.3. Research questions

- RQ1: How much does multi-layer filtering improve downstream extraction reliability?
- RQ2: Which merit-classification model is most robust under class imbalance?
- RQ3: How do segmentation quality and NER quality interact in the final extraction output?

1.4. Main contributions

1. End-to-end framework from raw legal documents to structured entities.
2. Task-adaptive preprocessing design with separate outputs for classification and NER.
3. Comparative evaluation of embedding-based classifiers for penal-merit prediction.
4. Practical protocol for LLM-assisted segmentation plus legal-domain NER.

2. Related Work

The proposed pipeline spans four distinct technical areas: legal document classification and filtering, sentence segmentation and structural analysis of legal texts, named entity recognition in legal and low-resource language settings, and end-to-end legal NLP frameworks. This section surveys the most relevant prior work in each area, identifying the key contributions and limitations that motivate the design choices made in the present study.

2.1. Legal document filtering and classification

The ability to distinguish legally relevant documents from heterogeneous corpora is a prerequisite for any downstream extraction task. Ariai et al. [02_02] provide a comprehensive survey of NLP tasks in the legal domain, systematically characterizing the unique challenges that complicate classification: extensive document lengths, highly specialized vocabulary, jurisdiction-dependent terminology, and the scarcity of openly available annotated datasets. The authors identify Legal Text Classification (LTC) as a well-studied but unevenly distributed subfield, with a strong concentration of work on Chinese and English legal systems

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and comparatively little attention to other languages and jurisdictions. Their identification of domain-specific fine-tuning as a critical success factor directly informs the use of Portuguese Legal-BERT and BERTimbau encoders in Stage2 of the present work.

Resck et al. [02_01] demonstrate the effectiveness of combining a transformer-based classifier with an interpretability mechanism for the identification of implicit citation relationships in Brazilian Supreme Court decisions. Although their task differs from multi-class merit classification, their architecture illustrates the viability of applying machine learning classification to large-scale Brazilian legal corpora — precisely the linguistic and institutional context addressed in the present study. The visual analytics layer they develop also highlights the practical importance of explainability in legal AI systems. Nuranti et al. [02_03] contribute a systematic, reproducible pipeline for corpus construction and NER annotation covering Indonesian judicial decisions across three hierarchical court levels. Their use of BIO tagging with more than 50 entity types, combined with Fleiss' Kappa-based inter-annotator agreement evaluation, establishes methodological best practices that inform the annotation and quality-control protocol adopted in the present work.

2.2. Legal sentence segmentation and structural analysis

Accurate segmentation of legal documents into semantically meaningful units is a precondition for reliable information extraction. Bhattacharya et al. [03_05] present DeepRhole, a deep learning system for rhetorical role labeling of sentences in Indian and UK Supreme Court case documents, assigning functional labels — such as Fact, Argument, Precedent, and Final Judgement — to individual sentences. Their experimental evaluation demonstrates that Hierarchical BiLSTM architectures substantially outperform prior CRF-based methods that relied on handcrafted features, and that domain-specific pretraining using Law2Vec consistently improves performance. Crucially, they also report that some rhetorical label distinctions (e.g., Ratio of the Decision vs. Precedent) are inherently subjective, a finding that has direct methodological implications for any segmentation protocol that must reconcile automated predictions with human expert judgment.

Sheik et al. [03_07] address sentence boundary detection (SBD) in English legal texts, demonstrating that a hybrid CNN+CRF architecture outperforms both standalone deep learning models and open-source SBD libraries, achieving more than 4% F1 improvement. Their out-of-domain evaluation on Indian legal text confirms the generalizability of context-aware delimiter classification to unseen legal corpora — a relevant consideration for the present work, where the Brazilian Portuguese legal register introduces additional boundary ambiguity through citation strings, procedural enumerations, and multi-clause sentence structures. Almalki et al. [03_03] propose a segmentation-and-aggregation framework for legal judgment prediction in Arabic, in which documents are segmented into sentence-level inputs, independently classified with AraBERT, and then aggregated to produce document-level decisions. Their success in overcoming the 512-token transformer limit through sentence-level decomposition provides a structural precedent for the LLM-assisted segmentation approach adopted in Stage3 of the present pipeline.

Ragazzi et al. [03_04] introduce LAWSUIT, a large-scale dataset for abstractive summarization of Italian constitutional court verdicts, and demonstrate that segmentation-based summarization pipelines outperform end-to-end sequence-to-sequence models across both full and few-shot settings. Their finding — that the uniform distribution of salient content throughout Italian legal verdicts prevents position-based shortcuts — resonates with the challenge encountered in Brazilian criminal labor-law decisions, where legally critical information (the conviction, the penalty, and the parties' identities) is similarly distributed across structurally heterogeneous document sections. Chen et al. [03_06] independently confirm the superiority of hierarchical segmentation in the context of long-document summarization using LLMs, showing that combining document decomposition with Chain-of-Thought (CoT) prompting significantly reduces factual hallucination and improves coverage across scientific, governmental, and legal domains. Their ablation studies confirm that segmentation and reasoning guidance make independent, complementary contributions to output quality.

2.3. Named entity recognition in legal and low-resource settings

Legal NER is the most direct antecedent to Stage4 of the present pipeline. Wang et al. [03_02] propose AT-Flat, an adversarial training model that integrates task-shared word boundary signals from Chinese Word Segmentation (CWS) into a nested NER architecture for Chinese judicial documents. By combining character-word fusion via the Flat-Lattice Transformer with an adversarial multi-task framework, they address the nested entity structure that is also present in Brazilian legal text, where party names, judicial organizations, and statutory references frequently overlap. Their demonstration that adversarial boundary transfer improves NER performance in specialized legal corpora — evaluated on traffic accident and financial lending judgment datasets — validates the importance of domain-aware architectural design for legal NER.

Bellandi et al. [01_01] present an entity-centric infrastructure for Italian court judgments that combines machine learning and syntactic rule-based NLP services in an iterative annotation pipeline with a human-in-the-loop validation layer. Their co-design methodology, involving legal institutions and implementation partners alongside a scientific team, and their emphasis on building a sustainable, multi-purpose annotation infrastructure rather than a single-task

system, closely mirrors the design philosophy of the present framework. The scalability challenges they identify — particularly the difficulty of maintaining annotation quality at national scale with limited expert involvement — are directly relevant to the Brazilian context, where the volume of labor-court decisions is substantial and expert annotation resources are scarce. Sakhaee and Wilson [01_02] demonstrate that information extraction approaches based on rule-based identification of citation expressions can achieve precision and recall exceeding 98% when constructing legislation networks from large, noisy legal corpora. Their success in handling poor-quality scanned documents through approximate string matching underscores the practical value of preprocessing robustness — a principle instantiated in Stage1 of the present pipeline through aggressive PJe footer removal and document-type normalization.

Wang et al. [03_09] propose CLWKD, a cross-lingual semantic alignment framework that leverages multi-granularity mapping matrices and knowledge distillation to transfer domain-specific representations from high-resource languages to low-resource agglutinative languages. While developed for Mongolian-Chinese and Korean-Chinese NLP tasks, the framework's motivation — overcoming the joint challenges of domain-specific data scarcity and morphological complexity — is structurally analogous to the resource constraints faced in Brazilian Portuguese legal NLP, where both domain-adapted models and annotated legal corpora remain limited relative to English counterparts. The paper establishes cross-lingual knowledge distillation as a promising paradigm for extending legal NLP capabilities to underrepresented languages without requiring large-scale domain-specific annotated corpora.

Zhu et al. [01_03] design a knowledge graph retrieval system for Chinese public prosecutorial affairs, combining a bi-LSTM-CRF NER module with ontology-based graph storage and latent semantic indexing. Their multichannel input strategy — fusing text representations, word vectors, and entity vectors — and their integration of NER within a broader knowledge management infrastructure provide a relevant reference for the entity aggregation and output schema design in Stage4 of the present work. Similarly, Guha et al. [03_08] demonstrate that multi-modal document classification — combining VGG16 image features with Legal-BERTbase text representations — achieves F1-scores exceeding 97% for document boundary detection in a Title Insurance context. Their confirmation that textual features are the primary determinant of classification performance, with visual features providing marginal but operationally significant gains, reinforces the text-centric design of the present pipeline.

2.4. Comparative synthesis

Table 1 summarizes the most closely related studies, positioning each work with respect to the dataset, language, primary task, model family, and reported metrics. The comparison highlights the principal research gap addressed by the present study: no prior work has proposed a multi-layer, end-to-end legal NLP framework for Portuguese-language criminal labor-law decisions that jointly addresses document-type filtering, penal-merit classification, structured segmentation, and named-entity extraction within a single, cohesive pipeline. Existing pipelines are either single-stage, language-specific (predominantly English or Chinese), or focused on a subset of the tasks addressed here.

3. Dataset and Problem Definition

3.1. Data acquisition and corpus description

The corpus used in this study was collected from the public consultation portal of the *Processo Judicial Eletrônico* (PJe) system maintained by the Federal Regional Court of the 1st Region (TRF1)[1]. A purpose-built web scraper, implemented in Python with the Selenium browser-automation library, was used to programmatically retrieve documents associated with criminal proceedings filed under Article 149 of the Brazilian Penal Code (*redução a condição análoga à de escravo*). For each process number listed in a pre-compiled spreadsheet, the scraper navigates the PJe search interface, opens the corresponding case page, and downloads every available PDF attachment. The raw PDF files are then converted to plain text via optical character recognition (OCR), producing the textual content that feeds the subsequent pipeline stages.

The resulting raw corpus comprises 2,581 documents drawn from 565 unique criminal proceedings spanning the period 2010–2024, with the bulk of the documents concentrated between 2018 and 2023. The cases originate from federal courts across multiple Brazilian states, including Goiás, Pará, Mato Grosso, Maranhão, Piauí, Bahia, Amapá, and the Federal District. Each record is stored with two fields: the source PDF filename (which encodes the process number) and the extracted full text. The first line of each document's text carries a PJe-generated header of the form `Documento id <N> - <Type>`, which provides the ground-truth document type.

The corpus exhibits substantial heterogeneity. Table 2 summarises the distribution of the main document types. Procedural dispatches (*despachos*, $n = 923$) and interlocutory decisions (*decisões*, $n = 659$) together account for more than 61% of the corpus; administrative records (*atos ordinátorios*, $n \approx 510$) and hearing minutes (*atas de audiência*, $n = 219$) make up most of the remainder. Only 99 documents are judicial sentences (*sentenças*), distributed across four PJe sub-types (Tipo D: 69; Tipo E: 17; Tipo A: 10; Tipo C: 3), and 16 are appellate rulings (*acórdãos*). This pronounced scarcity of decision-bearing documents—less than 4.5% of the total—motivates the multi-layer filtering strategy described in Section 4.

A preliminary quality inspection revealed pervasive noise introduced by the PJe platform itself: duplicate footer blocks containing page identifiers, electronic-signature attestations, public-consultation URLs, and internal document

[1] https://pje1g.trf1.jus.br/consultapublica/

Table 1
Comparative summary of related works. SBD = Sentence Boundary Detection; RRL = Rhetorical Role Labeling; LJP = Legal Judgment Prediction; NER = Named Entity Recognition; AS = Abstractive Summarization; DSS = Document Stream Segmentation; F1 = macro/weighted F1-score; Acc = accuracy.

Study	Ref.	Lang.	Task	Model family	Metric	Key limitation
Bellandi et al. (2024)	[01_01]	IT	NER / Annotation framework	ML + rule-based NLP	Feasibility	Single language; limited labelled data
Sakhaee & Wilson (2021)	[01_02]	EN	Citation extraction / legislation network	Rule-based IE	98% P&R	No learning; specific to NZ legislation
Zhu et al. (2022)	[01_03]	ZH	NER / Knowledge graph	BiLSTM-CRF + Neo4j	Acc (crime pred.)	Single domain; Chinese only
Resck et al. (2023)	[02_01]	PT	Citation classification	ML + interpretability	96% Acc	Single court; binary task
Ariai et al. (2025)	[02_02]	Multi	Survey (LTC, NER, LJP, etc.)	Survey	—	Survey; no new experiments
Nuranti et al. (2025)	[02_03]	ID	NER corpus construction	BiLSTM-CRF / BIO annotation	IAA 0.705	Method paper; no trained model reported
Wang et al. (2025)	[03_02]	ZH	Nested NER in legal text	AT-Flat (FLAT + adversarial)	State-of-the-art F1	Chinese only; long-doc limitations
Almalki et al. (2025)	[03_03]	AR	LJP / segmentation-aggregation	AraBERT + aggregation	85.62% Acc	Binary outcome; Arabic only
Ragazzi et al. (2025)	[03_04]	IT	Legal abstractive summarization	Seq2seq + segmentation	ROUGE	Summarization only; Italian only
Bhattacharya et al. (2023)	[03_05]	EN	Rhetorical role labeling	Hier-BiLSTM / BERT / Legal-BERT	F1 0.82 (IN)	EN/IN focus; no PT variant
Chen et al. (2025)	[03_06]	Multi	Long-doc summarization	CoT + hierarchical LLM	ROUGE / FactCC	High compute; summarization only
Sheik et al. (2025)	[03_07]	EN	Sentence boundary detection	CNN + CRF hybrid	F1 +4%	English legal text; no multilingual eval
Guha et al. (2022)	[03_08]	EN	Document stream segmentation	VGG16 + Legal-BERT (multimodal)	F1 97.37%	Proprietary data; TI domain only
Wang et al. (2026)	[03_09]	ZH/MN/KO	Cross-lingual domain alignment	CLWKD (mapping + KD)	Cross-lingual NLP	Low-resource focus; no legal-specific eval
This work	—	**PT**	**Filtering + classification + segmentation + NER**	**Legal-BERT + classical + LLM + NER**	**F1 / pipeline**	**Low-resource; single court system**

numbers are repeated throughout the extracted text of nearly every document. These artefacts, if left untreated, would degrade both classification and entity-extraction performance; their systematic removal is therefore addressed in the preprocessing stage (Section 4.3).

3.2. Task definitions

The framework decomposes the end-to-end information-extraction problem into four sequential tasks, each with a clearly defined input, output, and evaluation criterion.

T1: Document-type filtering. Given the raw heterogeneous corpus ($N = 2{,}581$), assign each document to one of seven canonical types and retain only judicial sentences (*sentenças*). This is a deterministic classification task whose correctness can be verified exhaustively because the ground-truth label is embedded in the PJe header. The expected output is a filtered subset of $n = 99$ sentences suitable for merit analysis.

T2: Penal-merit classification. Given a preprocessed sentence text, predict its penal-merit outcome among three mutually exclusive classes: *condenação* (conviction, $n = 25$), *absolvição* (acquittal, $n = 46$), or *extinto* (extinguished criminal liability, $n = 18$). Documents whose merit labels

Table 2
Distribution of the main document types in the raw corpus ($N = 2{,}581$).

Document type	Count
Despacho	923
Decisão	659
Ato ordinátorio	510
Ata de Audiência	219
Sentença (all sub-types)	99
Certidão de Trânsito em Julgado	81
Decisão Monocrática Terminativa	34
Voto	19
Acórdão	16
Certidão de Julgamento	18

fall outside these three categories—such as appellate reviews, partial rulings, or procedurally dismissed cases—are excluded prior to classification. The task is evaluated via stratified k-fold cross-validation using weighted F1 as the primary metric (Section 4.4).

T3: Sentence-level segmentation. Given a conviction-class decision ($n = 25$), decompose it into individual sentences and assign each sentence a rhetorical-function label drawn from a closed set of six categories: RELATÓRIO (case summary), DOS_FATOS (factual narrative), FUNDAMENTAÇÃO (legal reasoning), DOSIMETRIA (penalty calculation), DISPOSITIVO (operative ruling), and OUTROS (non-substantive content). The segmentation must be exhaustive—every token in the source document must be covered by exactly one labelled segment—and the output must conform to a predefined JSON schema to enable automated downstream consumption. Evaluation criteria include schema-validity rate, label-distribution plausibility, and expert agreement on a reviewed subset.

T4: Named-entity extraction. Given a segmented and labelled sentence, identify and classify legally relevant named entities using a domain-specific NER model. Target entity types include person names, organizations, locations, dates, monetary values, statutory references, and crime descriptions. The task operates over the segmented output of T3, allowing entity extraction to benefit from the structural context provided by rhetorical labels—for example, party names are expected predominantly in RELATÓRIO segments, while penalty amounts concentrate in DOSIMETRIA. Extracted entities are aggregated into a structured JSON record per document, designed for downstream integration with external databases such as Brazil's *Lista Suja* of employers found responsible for modern-slavery conditions.

3.3. Annotation and label governance

Label quality is a critical concern throughout the pipeline, given the specialized legal knowledge required to interpret criminal labor-law decisions and the absence of a pre-existing annotated benchmark for this domain. Each task relies on a distinct labelling strategy, reflecting the nature of the label and the feasibility of automated versus expert-driven annotation.

T1 labels (document type). Document-type labels are extracted directly from the PJe system header and therefore require no manual annotation. Because the label is machine-generated by the court's electronic filing system, it functions as a *de facto* ground truth. A random sample of 50 documents was manually inspected to confirm that the extracted type matched the document content; no discrepancies were found.

T2 labels (penal merit). Merit labels (*condenação*, *absolvição*, *extinto*) were assigned by a legal-domain expert who read each of the 99 filtered sentences in full. The annotation criterion is whether the decision examines the criminal liability of the defendant under Article 149 of the Brazilian Penal Code and, if so, what the outcome is. Convictions include cases of partial conviction—i.e., when at least one defendant was found guilty on the Article 149 charge, even if co-defendants were acquitted or charges under other statutes were dismissed. The *extinto* class covers cases in which criminal liability was formally extinguished (typically due to statute-of-limitations expiry) before a substantive merit ruling was reached. Documents that did not fit any of the three categories—such as decisions on procedural motions, *embargos de declaração*, or interlocutory appeals that had been misclassified as sentences by the PJe header—were removed from the classification dataset, reducing the working set from 99 to 89 documents.

Due to resource constraints, a single annotator performed the merit labelling. To mitigate the risk of systematic bias, a second reviewer (a co-author with legal training) independently classified a stratified subset of 30 documents (approximately one-third of the corpus, sampled proportionally across the three classes). Disagreements were resolved through discussion, and the final labels reflect the consensus outcome. While a full inter-annotator agreement study with multiple independent annotators would strengthen the evidence, the combination of expert annotation with partial double-coding and consensus resolution represents a pragmatic quality-assurance protocol for a low-resource setting.

T3 labels (segmentation). Segmentation labels are produced by a large language model (Gemini 2.5 Flash Lite) operating under a detailed annotation prompt that specifies six rhetorical-function categories and 21 decision rules governing label assignment (see Section 4.3 and the supplementary material for the full prompt). The prompt was iteratively refined through pilot runs on five documents, with a legal-domain expert reviewing the output at each iteration to identify systematic errors and ambiguous boundary cases. The final prompt version was then applied to the full conviction subset ($n = 25$), of which 17 documents were successfully processed and 8 returned parsing errors due to malformed JSON or context-length overflow.

Following LLM annotation, the 17 successfully segmented documents were reviewed by the legal expert, who verified label correctness at the sentence level and flagged

cases where the model had assigned an incorrect rhetorical role. This human-in-the-loop validation step serves as a lightweight surrogate for a formal gold standard. We acknowledge that the current protocol does not include a quantitative inter-annotator agreement metric (e.g., Cohen's κ) for segmentation; establishing such a benchmark with multiple annotators is planned as future work.

T4 labels (NER). Named-entity extraction relies on the pre-trained model `dominguesm/legal-bert-ner-base-cased-ptbr`, which was trained on a Portuguese legal NER corpus with BIO-tagged entity types. No additional annotation was performed for this stage; evaluation is based on the model's intrinsic predictions over the segmented conviction documents. A formal entity-level evaluation against a manually annotated gold standard is deferred to future work, pending the construction of a schema-validated reference dataset for at least 15–20 documents.

3.4. Data governance and ethics

All documents used in this study are judicial decisions issued by Brazilian federal courts and made publicly available through the PJe electronic consultation portal. Under Brazilian law, criminal proceedings are, as a general rule, public (*princípio da publicidade*, Article 93, IX of the Federal Constitution), and the decisions analysed here were not subject to judicial secrecy (*segredo de justiça*) at the time of collection. The corpus was therefore assembled exclusively from publicly accessible records, and no privileged or restricted-access data was used at any stage.

Privacy and data protection. Although the source documents are public, they contain personal data—including the full names of defendants, victims, witnesses, judges, and attorneys—that falls within the scope of Brazil's General Data Protection Law (Lei Geral de Proteção de Dados, LGPD, Law 13.709/2018). The present study processes these data for academic research purposes, which constitutes a legitimate basis under Article 7, IV of the LGPD. No personally identifiable information is disclosed beyond what is already contained in the publicly available court records. The structured outputs produced by the NER stage (T4) do extract person names and organizational identifiers; however, these outputs are stored locally and are not published as part of this paper's supplementary material. Any future public release of the annotated dataset will apply pseudonymization to defendant and victim names, replacing them with anonymized identifiers while preserving the relational structure needed for supply-chain analysis.

Ethical considerations. The research does not involve human subjects in the experimental sense—no interviews, surveys, or behavioural observations were conducted. The annotation tasks (T2 merit labelling and T3 segmentation review) were performed by members of the research team and did not involve external crowd workers or vulnerable populations. The subject matter—criminal convictions for modern slavery under Article 149 of the Brazilian Penal Code—is inherently sensitive, as it involves severe labour exploitation. We note that the framework is designed to support transparency and accountability in supply-chain governance by enabling the systematic extraction of structured information from court records. It is not intended, and should not be used, to make automated legal determinations or to replace judicial decision-making processes.

Institutional oversight. The study was conducted under the academic supervision of the Federal University of Rio Grande do Sul (UFRGS). As the research relies exclusively on publicly available court records and does not involve direct interaction with human participants, formal ethics committee approval (Comitê de Ética em Pesquisa) was not required under the applicable Brazilian regulations (Resolution CNS 510/2016, Article 1, sole paragraph, items III and V). Nonetheless, the data-handling procedures were designed to comply with LGPD requirements and institutional data-governance policies.

4. Methodology

4.1. Overview of the multi-layer pipeline

Figure 1 presents the end-to-end workflow adopted in this study. The process starts with document-type normalization and filtering (Stage1_1), keeping only judicial sentences that are relevant to criminal-merit analysis. Next, Stage1_2 applies task-adaptive preprocessing and generates two normalized outputs: one optimized for merit classification and one that preserves legal-text patterns for segmentation and NER. Stage2 performs penal-merit classification, Stage3 applies LLM-assisted segmentation to the conviction subset, and Stage4 performs legal-domain named-entity extraction over segmented content.

4.2. Stage1_1: Document-type normalization and filtering

As noted in Section 3.1, every document in the PJe system carries a machine-generated header line of the form `Documento id <N> - <Type>`. Stage1_1 exploits this metadata to perform rule-based document-type classification without requiring any trained model. A regular expression extracts the type string from the first line of each document; the extracted label is then mapped to one of seven canonical categories—*Sentença*, *Decisão*, *Despacho*, *Ata de Audiência*, *Ato Ordinátorio*, *Acórdão*, or *Outro*—via case-insensitive substring matching. This deterministic approach is feasible because the type label is generated by the court system itself and is therefore highly reliable.

After classification, only documents labelled *Sentença* are retained for subsequent processing. This single filtering step reduces the corpus from 2,581 to 99 documents—a 96.2% reduction—thereby removing all procedural dispatches, interlocutory decisions, hearing minutes, and other ancillary records that do not contain merit-level adjudications. Acórdãos (appellate rulings, $n = 16$) are also excluded at this stage because they typically re-evaluate a prior sentence rather than introduce new factual findings, and their

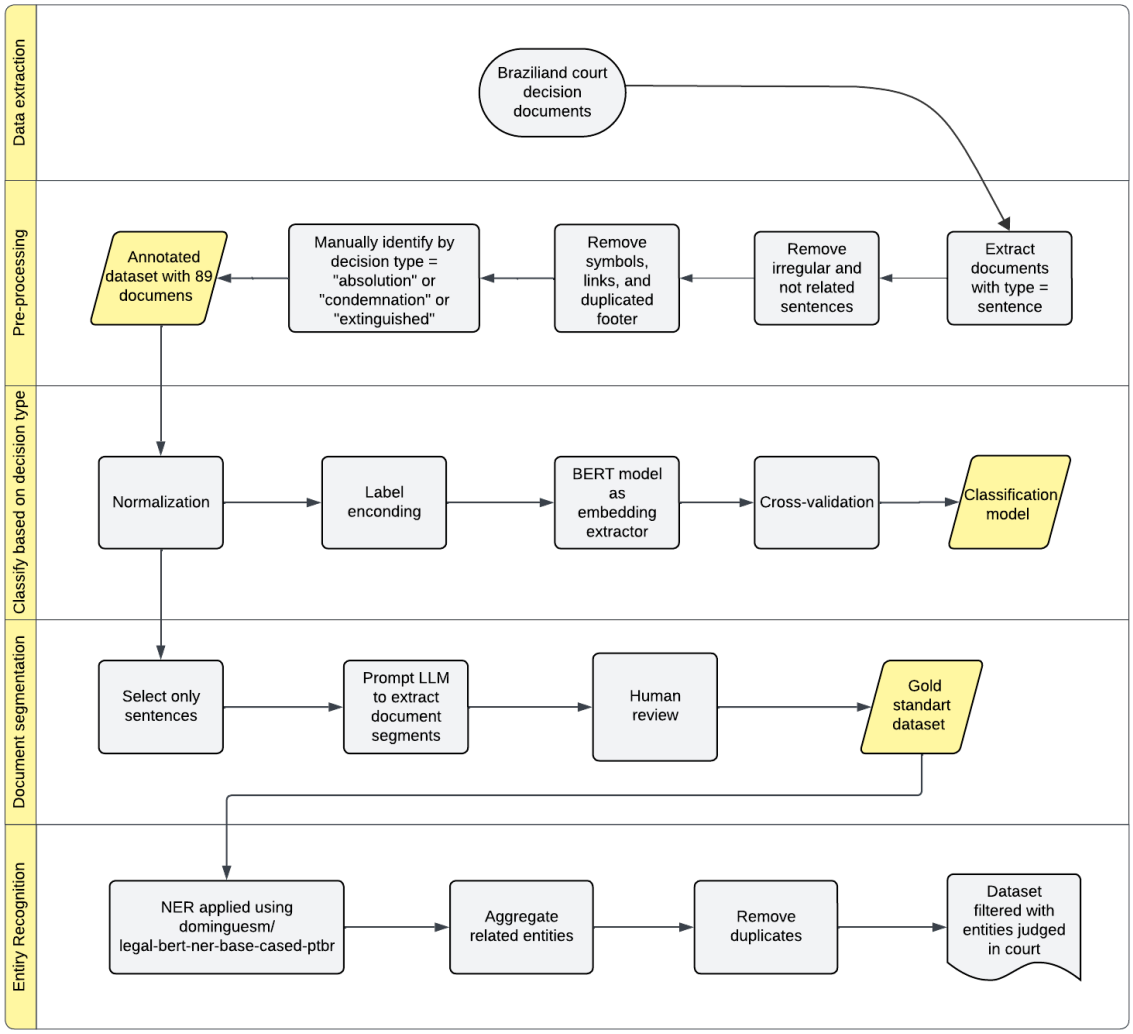

Figure 1: Activity diagram of the proposed multi-layer pipeline for legal document filtering, penal-merit classification, segmentation, and NER extraction.

inclusion would risk duplicating information already present in the corresponding first-instance sentence.

4.3. Stage1_2: Task-adaptive legal text preprocessing

The 99 sentences retained by Stage1_1 still contain substantial noise inherited from the PJe platform. Stage1_2 addresses this through a shared cleaning step followed by two divergent normalization branches, each tailored to a different downstream task.

Shared step: PJe footer removal. Every page of a PJe document is stamped with a set of pseudo-lines: page identifiers (`Num. . . . - Pág. . . .`), electronic-signature attestations, public-consultation URLs, and internal document-number references. Because these lines are repeated on every page, they can constitute a significant fraction of the extracted text and introduce misleading token frequencies. Six regular expressions, applied sequentially, strip all such footer variants; runs of excessive whitespace are then collapsed.

Classification branch (aggressive normalization). The first output targets the downstream merit classifier (Stage2), which must distinguish between *condenação*, *absolvição*, and *extinto* based on semantic content rather than surface cues. Accordingly, this branch removes: (i) formulaic legal phrases (*e.g.*, "Vistos, etc.", "Passo a decidir", "Publique-se, registre-se, intimem-se"); (ii) structured identifiers such as CNJ case numbers, long digit sequences, and monetary values; (iii) party names rendered in ALL-CAPS; (iv) procedural role keywords (*e.g.*, autor, réu, apelante); (v) dates in both numeric and spelled-out formats; and (vi) statutory references (article, paragraph, and subsection numbers). Finally, the text is lowercased and Unicode-normalized (NFKC). A safety guard restores the pre-normalization text whenever the aggressive pipeline produces a result shorter than 30 characters.

NER/segmentation branch (conservative normalization). The second output is designed for the segmentation (Stage3) and named-entity recognition (Stage4) tasks, which rely on the very tokens that the classification branch discards—party names, dates, monetary values, legal references, and structural markers. This branch therefore applies only Unicode normalization and whitespace regularization (collapsing horizontal runs and excessive blank lines) while preserving the original casing and all substantive content.

The dual-output design allows each downstream stage to receive text in which irrelevant variation has been reduced without sacrificing the features it needs, a strategy that avoids the common trade-off between classification-oriented cleaning and entity-preserving fidelity.

4.4. Stage2: Penal-merit classification with BERT embeddings

The goal of Stage2 is to classify each sentence into one of three penal-merit categories: *condenação* (conviction, $n = 25$), *absolvição* (acquittal, $n = 46$), or *extinto* (extinguished liability, $n = 18$). After discarding documents with labels outside these three classes and those with fewer than 10 characters of normalized text, the working set comprises 89 documents—a low-resource setting that constrains the choice of modelling strategy.

Embedding generation. Rather than fine-tuning a transformer end-to-end—an approach we tested but discarded due to high variance on this sample size—we adopt a feature-extraction strategy in which a pre-trained BERT encoder produces a fixed-dimensional sentence embedding for each document, and a classical supervised classifier is trained on top. Three encoder backbones are compared:

1. `dominguesm/legal-bert-base-cased-ptbr`, a Portuguese legal-domain BERT;
2. `neuralmind/bert-base-portuguese-cased` (BERTimbau base), a general-purpose Portuguese BERT; and
3. `neuralmind/bert-large-portuguese-cased` (BERTimbau large), the corresponding large variant.

All encoders are loaded via the `SentenceTransformer` library. For each cross-validation fold, the encoder maps every document in the training and test partitions to a dense vector; a `StandardScaler` is then fitted on the training embeddings and applied to both sets before classification.

Classifiers. Four classifiers are evaluated on the scaled embeddings:

- Logistic Regression (`max_iter`= 1000, `class_weight=balanced`);
- Linear SVM (`class_weight=balanced`);
- Random Forest (150 estimators, `max_depth`= 6, `class_weight=balanced`); and
- XGBoost (150 estimators, `max_depth`= 4, `learning_rate`= 0.1).

Class-weight balancing is applied wherever supported to mitigate the effect of the skewed label distribution.

Evaluation protocol. All classifiers are evaluated under stratified k-fold cross-validation with $k = 3$ (automatically reduced if the minority class has fewer samples than k), `shuffle=True`, and a fixed random seed (42). Per-fold metrics include accuracy, weighted precision, weighted recall, and weighted F1; aggregate performance is reported as the mean and standard deviation of the weighted F1 across folds. In addition, predictions from all folds are pooled to produce a single classification report with per-class scores and a confusion matrix, enabling a more granular view of class-level behaviour.

4.5. Stage3: LLM-assisted segmentation of conviction decisions

- Input restricted to conviction subset.
- Prompting protocol and JSON schema validation.
- Retry/failure handling and quality-control checks.

4.6. Stage4: Legal-domain NER extraction

- Model: `dominguesm/legal-bert-ner-base-cased-ptbr`.
- Entity aggregation strategy and output schema.

5. Experimental Setup

5.1. Evaluation protocol

- Data splits, fold strategy, seeds, and confidence intervals.
- Primary metrics: macro-F1, weighted-F1, precision, recall, per-class scores.

5.2. Baselines and ablations

- Baseline without multi-layer filtering.
- Baseline with single normalization stream.
- Ablation removing each stage (one-at-a-time).

5.3. Implementation details

- Software stack, hardware profile, runtime, and cost profile (LLM calls).

6. Results

6.1. Merit classification performance

Table 3 compares three encoder backbones under the same downstream classifiers, using weighted F1 as the primary criterion. The best overall result is highlighted in bold and corresponds to BERTimbau-large with linear SVM ($F1 = 0.6152 \pm 0.0808$). Legal-BERT remains competitive and obtains the strongest XGBoost result ($F1 = 0.6114 \pm 0.0871$), while BERTimbau-base shows gains for Logistic Regression and Random Forest.

In practical terms, this indicates that domain adaptation in the encoder remains beneficial for the XGBoost

Table 3
Stage2 encoder comparison across downstream classifiers (weighted F1, mean ± std across folds).

Classifier	Legal-BERT (dominguesm)	BERTimbau-base (neuralmind)	BERTimbau-large (neuralmind)
XGBoost	0.6114 ± 0.0871	0.5261 ± 0.0590	0.5661 ± 0.0558
SVM (Linear)	0.5415 ± 0.0484	0.5350 ± 0.0664	$\mathbf{0.6152 \pm 0.0808}$
Random Forest	0.5406 ± 0.1249	0.5611 ± 0.0622	0.5284 ± 0.0420
Logistic Regression	0.5399 ± 0.0522	0.5627 ± 0.0613	0.6136 ± 0.0799

configuration, while larger general-purpose Portuguese encoders can provide stronger performance with linear classifiers. Given this trade-off, both Legal-BERT+XGBoost and BERTimbau-large+SVM should be considered viable Stage2 operating points depending on interpretability and computational constraints.

TODO: Mention that the BERT models were not fine-tuned for this specific case due to the limited amount of available data.

6.2. Segmentation quality

Report schema-valid rate, section-level agreement, and representative errors.

6.3. NER extraction quality

Report entity-level precision/recall/F1 (strict and relaxed matching if possible).

6.4. End-to-end pipeline gains

Compare extraction quality before vs. after multi-layer filtering.

7. Discussion

7.1. Why the pipeline works
7.2. Failure modes and error taxonomy
7.3. Generalization to other legal domains/languages
7.4. Operational and policy implications

8. Threats to Validity

- Internal validity: label noise, annotation drift, prompt sensitivity.
- External validity: transfer to other courts, years, and jurisdictions.
- Construct validity: legal meaning vs. NLP proxy metrics.

9. Reproducibility Statement

- Public code structure, dependency pinning, random seeds, and artifact inventory.
- Which resources can/cannot be publicly shared due to legal constraints.

10. Conclusion and Future Work

- Summarize empirical and practical contributions.
- Future work: active learning, expert-in-the-loop verification, and cross-court transfer evaluation.

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CRediT authorship contribution statement

Carlos A. V. Tinin: Conceptualization, Methodology, Software, Writing – original draft. **Edison P. de Freitas:** Validation, Formal analysis, Writing – review & editing. **Kevin H. M. Gularte:** Validation, Formal analysis, Writing – review & editing. **Elena J. da Costa:** Validation, Formal analysis, Writing – review & editing. **João Paulo J. da Costa:** Validation, Formal analysis, Writing – review & editing.

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